ECE 329 **Homework 6** **Due: Fri, Nov 7, 2025, 11:59PM**

6 problems totaling 60pts.

1. (10pts)

   a) (3pts) I have a slab of some insulator with an unknown permittivity ϵ. To determine ϵ experimentally I go to the lab and insert the slab in between the plates of a capacitor whose plate spacing exactly matches the width of the slab. I observe that the time constant of exponential decay of the capacitor voltage in an RC circuit that I construct increases by 25% when the slab is inserted to replace the air spacing. Determine ϵ in terms of ϵ_o. Explain your reasoning carefully.

   b) (4pts) Repeat (a) if the slab width is only one half of the plate separation so that when the slab is inserted between the plates we still have 50% air filling. **Hint: D** will be the same in both dielectrics but not **E**, and, hence, capacitor voltage $V = \int_{top}^{bottom} \mathbf{E} \cdot d\mathbf{l}$ will be influenced!

   c) (3pts) I have a rod of some solid with an unknown permeability μ. To determine μ experimentally I go to the lab and insert the rod within a solenoid of many turns having a diameter exactly matching the diameter of the rod. I observe that the time constant of exponential decay of the solenoid current in an RL circuit that I construct increases by 0.05% when the rod is inserted to replace the air core of the solenoid. Determine μ in terms of μ_o. Is the rod diamagnetic ($\chi_m < 0$) or paramagnetic ($\chi_m > 0$)? Explain. **Hint**: the permeability $\mu = \mu_r \mu_0 = (1 + \chi_m)\, \mu_0$ where χ_m is the magnetic susceptibility of the material.

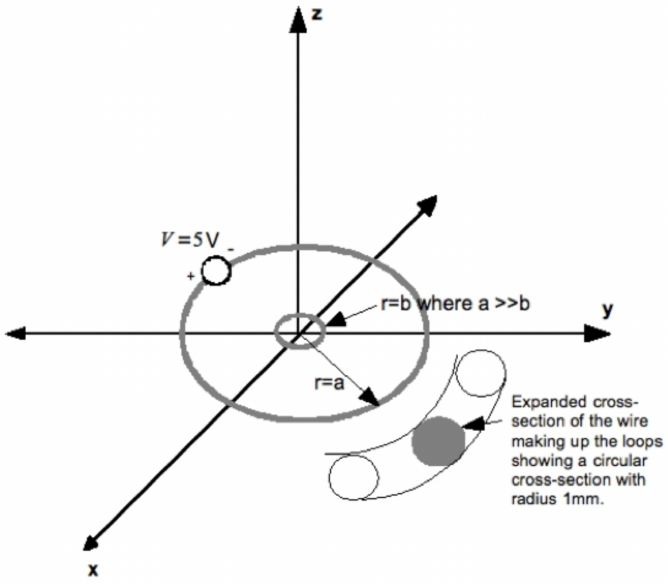
2. (5pts) For current density $\mathbf{J} = (3z^2\,\hat{x} + 2x^3y\,\hat{y} + z(y-y_o)^2\,\hat{z})$ A/m^2, which is time independent, find the charge density $\rho(0,t)$ at the origin (0,0,0) as a function of time t, if $\rho = 0$ at that location and time $t = 0$, given that $y_o = 1$ m. **Hint:** use the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

3. (10pts) Derive the Biot-Savart law from the Ampere's law and the Gauss' Law for static magnetic fields.

4. (10pts) Compute the magnetic vector potential $\mathbf{A}$ (5pts) and the magnetic fields $\mathbf{B}$ (5pts) along the z-axis from a circular loop of radius a carrying DC current I. The current loop is placed on the xOy plane centered at the origin. The current flows clockwisely looking along $\hat{z}$.

5. (15pts) Consider two concentric circular wire loops of radii $a = 1$ m and $b = 0.02$ m placed on the xOy plane of the reference coordinate system with their centers at the origin. The medium is free space. The conductivity of the wire from which both loops are made is $\sigma = 4 \times 10^7$ S/m. The cross-section of the wire is circular with radius 1 mm. A 5 V battery is connected in the outer loop (see figure).



a) (2pts) Calculate the current I_a that flows in the outer loop. **Hint:** first figure out the resistance for the loop given as $d/(A\sigma)$ where d is the length of the loop and A the cross sectional area of the loop wire.

b) (2pts) Derive an expression for the magnetic flux $\Psi_{a\to b}$ that links the inner loop due to the current flowing in the outer loop. Your expression should be in terms of the magnetic permeability of free space, the current I_a, and the radii of the two loops. **Hint:** Refer to the previous problem for expressions for the magnetic flux density due to a current flowing in a circular wire loop. Also, take advantage of the fact that $b \ll a$ so that the magnetic field across the smaller loop can be considered nearly constant.

c) (2pts) Derive the mutual inductance $L_{a\to b} \equiv \Psi_{a\to b}/I_a$ between the outer and inner loop.

d) (4pts) Assume next that the inner loop is moving upwards, in the positive z direction, with speed $v_z = 1$ m/s and center position $z = v_z t$. Calculate the *emf* induced in the inner loop at time t *assuming* that the flux linking the loop due to the induced current is negligible.
Hint: Neglecting self-inductance, $emf \approx \mathcal{E}_{a\to b} = -\frac{d\Psi_{a\to b}}{dt}$.

e) (5pts) You are given that the self-inductance of the inner loop is 0.25 μH. If the loop is moving upwards with speed $v_z = 1$ m/s, derive a differential equation for the induced current in the loop.

Hint: the current flowing around the inner loop is the induced emf of the loop divided by the loop resistance while the emf is the negative of the time derivative of the total magnetic flux produced by the currents flowing in both loops — constructing an expression for the current amounts to finding the differential equation that is requested.

6. (10pts) Consider an coaxial cable with inner radius a and outer radius b. The inner and outer conductors are both perfect electric conductors (PEC). The material filling the space ($a < r < b$) has relative permittivity $\varepsilon_r = 4$ and relative permeability $\mu_r = 2$. Assume the DC current is uniformly distributed over the cross-section of the inner conductor uniformly. Compute the per unit length inductance $\mathcal{L}$ of the cable. **Hint**: compute the associated magnetic energy per unit length and link it to the inductance.